Master Reference Document: Supervised Representation Learning for Multi-Omics Cancer Classification

A complete reference on my thesis project — current work, methodology, results, and future direction. Written to serve multiple purposes: explaining the work to a general audience, drafting SOP/CV content, and preparing for advisor meetings or presentations.

Quick-Reference Versions

One sentence (for a CV or LinkedIn): Converted an unsupervised multi-omics autoencoder into a supervised cancer-type classifier, achieving 99.46% test accuracy — competitive with Random Forest (99.73%) and outperforming SVM (94.89%) on TCGA pan-cancer data.

Two sentences (for a brief bio): My master's thesis, advised by Dr. Michael J. O'Connell (Miami University Department of Statistics), extends a published multi-omics autoencoder architecture (JISAE) by converting it from an unsupervised representation-learning model into an end-to-end supervised classifier for cancer type prediction. The work combines a structured literature review, full PyTorch reimplementation, and rigorous hyperparameter optimization to produce a model competitive with standard supervised baselines, while identifying open questions about the specific value added by the architecture's joint/individual embedding design.

CV bullet points:

- Reproduced and extended a published deep learning architecture (JISAE autoencoder) for multi-omics cancer genomics data in PyTorch
- Conducted structured literature review to identify established methods for converting unsupervised representation-learning models into supervised classifiers
- Designed and implemented a supervised classification head trained end-to-end via backpropagation through a pretrained encoder architecture
- Performed hyperparameter optimization using Optuna with 5-fold stratified cross-validation, achieving 99.46% test accuracy on a held-out TCGA pan-cancer dataset
- Benchmarked model performance against Random Forest and SVM baselines under an identical train/test protocol

One paragraph (for an SOP research-experience section): see the "Research Motivation and Context" section below, or the separate SOP narrative document.

Part 1: Understanding the Project (accessible explanation)

The problem in plain terms

Cancer researchers increasingly collect multiple types of molecular data from the same patients — gene expression, DNA methylation, microRNA levels — each offering a different, partial view of the same underlying biology. A central statistical challenge is combining these different data types into something useful and manageable, since each type alone can have tens of thousands of measurements per patient.

One popular tool for this is the **autoencoder**: a neural network that learns to compress high-dimensional data into a small "embedding," then tries to reconstruct the original data from that compression. If the reconstruction is accurate, the embedding must have captured the essential structure of the data.

The specific model at the center of my thesis, called **JISAE**, does this for two omics data types at once, and does something architecturally interesting: instead of just mixing both data types together into one embedding, it explicitly separates what is *shared* between the two data types from what is *unique* to each — producing three distinct compressed representations per patient rather than one.

The gap I'm addressing

JISAE, as originally built, is never told what cancer type a patient has while it's training — it only ever learns to compress and reconstruct. Classification is done afterward, as a separate step, by taking the frozen compressed representations and feeding them into standard classifiers like Random Forest or SVM. This works reasonably well, but not as well as simply feeding those classifiers the raw, uncompressed data directly.

My thesis asks: **what happens if we let the model learn directly from the cancer-type labels, rather than only being evaluated on them afterward?** This turns the model from *unsupervised* (learning without being told the "right answer") into *supervised* (learning directly toward a known goal), and tests whether that change is enough to make it competitive with standard supervised methods.

Why this matters beyond one dataset

The broader question is about representation learning generally: does forcing a compressed representation to preserve reconstruction-relevant information help or hurt when the real goal is prediction? And can architectural ideas designed for one training paradigm (unsupervised compression) be adapted to another (supervised prediction) without losing what made them useful in the first place? These questions apply well beyond cancer genomics, to any setting where high-dimensional, multi-source data needs to be compressed and used for prediction.

Part 2: The Foundational Work

The original paper

Wang, C., & O'Connell, M. J. (2025). *Autoencoders with shared and specific embeddings for multi-omics data integration.* BMC Bioinformatics, 26, 214.

This paper introduced JISAE (Joint and Individual Simultaneous Autoencoder) and compared it against several alternative autoencoder designs (CNC_AE, X_AE, MM_AE, MOCSS) and a classical statistical benchmark (JIVE — Joint and Individual Variation Explained), using both a simulation study and real TCGA pan-cancer data covering six cancer types (breast, lung, melanoma, liver, sarcoma, kidney) with paired gene expression (20,531 features) and microRNA expression (1,046 features) data, totaling 1,866 patients.

My reproduction work

Before extending the model, I independently reproduced JISAE in full — reimplementing the architecture, data preprocessing pipeline (including the exact train/test partitioning strategy), and training procedure in Python using PyTorch, and validating that my results matched the published findings. This required:

- Rebuilding the three-branch encoder architecture (gene expression branch, microRNA branch, and a joint branch processing the concatenation of both) producing three 512-dimensional embeddings (1,536 combined)
- Matching preprocessing decisions exactly (MinMaxScaler applied separately to train and test sets, 80/20 stratified split yielding 1,494 training and 372 test patients)
- Verifying my reconstruction loss and downstream classification accuracy matched the paper's reported numbers

This reproduction was not itself novel research, but was methodologically essential — genuine understanding of a published architecture requires rebuilding it, not just reading about it.

Part 3: Research Question and Literature Review

The research question

If JISAE's encoder is trained end-to-end with a classification objective — rather than trained purely for reconstruction and evaluated on classification only afterward — can it match or exceed the performance of standard supervised classifiers (Random Forest, SVM) on the same task?

Literature review

Before implementing any changes, I conducted a structured review to determine how other researchers have approached converting an unsupervised representation-learning model into a supervised one. This surfaced two distinct, established strategies:

Phased training approach:

- *OmiVAE* (Zhang et al., 2019, IEEE BIBM) — a variational autoencoder for pan-cancer classification, trained in two phases: unsupervised pretraining, followed by a supervised phase with a classifier attached. Reported 97.49% accuracy across 33 tumor types.
- *SubOmiEmbed* (Hashim et al., 2022) — extends this into three explicit phases: pretrain unsupervised, freeze the encoder and train only a classifier head, then unfreeze and fine-tune jointly.

Joint-loss training approach:

- *MOSAE* (Tan et al., 2020, BMC Medical Informatics and Decision Making) — trains reconstruction and classification simultaneously via a single combined loss (`L = L_reconstruction + λ·L_classification`). This traces back to theoretical work (Le, Patterson & White, 2018) arguing that reconstruction loss acts as a regularizer that improves generalization on the supervised task.

Benchmark reference (not implemented):

- *MOGONET* (Wang et al., 2021, Nature Communications) — a fully supervised multi-omics classifier built on graph convolutional networks. Architecturally unrelated to JISAE (it doesn't use an autoencoder at all), so not something to adopt directly, but a useful external benchmark for how a purpose-built supervised architecture performs on this class of problem.

This review confirmed that converting an autoencoder to a supervised classifier is an established, well-precedented transition rather than a novel or risky one — and gave a principled basis for choosing the phased/simpler approach as the starting point, with the joint-loss approach identified as a natural next experiment.

Part 4: Methodology

Design decision

Implemented the simpler of the two established approaches first: keep the encoder architecture completely unchanged, delete the decoder, and attach a small classification head trained via cross-entropy loss directly on cancer-type labels. This isolates a single, specific change — allowing the encoder's weights to be shaped by the classification objective — rather than combining multiple changes at once.

The four core code changes

1. **Output layer**: replaced the decoder's reconstruction output (dimension ≈21,577, matching the input size) with a small classification head ending in 6 outputs (one score per cancer type)
2. **Loss function**: replaced `MSELoss` (reconstruction error) with `CrossEntropyLoss` (classification error)
3. **Training target**: changed from the original input data to the integer-encoded cancer-type label
4. **Hyperparameter search objective**: changed Optuna's optimization direction from minimizing reconstruction loss to maximizing classification accuracy

Verification of the mechanism

Explicitly confirmed that gradients from the classification loss reach back into the encoder's earliest layer following a backward pass — verifying that the encoder genuinely learns from the classification signal, rather than only the newly added classifier head. This is the technical detail that distinguishes a genuinely end-to-end supervised model from one that is simply evaluated in a supervised manner after unsupervised training.

Hyperparameter optimization

Conducted a full re-optimization using Optuna (a Bayesian hyperparameter search framework), since the original hyperparameters were selected for a different objective (reconstruction) entirely:

- 5-fold stratified cross-validation, restricted entirely to the training partition (the 372-patient test set was never used for model selection)
- 50 trials, searching learning rate, weight decay, batch size, classifier head size, and dropout rate
- Early stopping per fold to avoid wasted computation on clearly underperforming configurations
- Best configuration achieved 99.53% average cross-validation accuracy, favoring a substantially smaller batch size and lower learning rate than the original reconstruction-tuned defaults — consistent with better-regularized, less aggressive training dynamics

Final evaluation protocol

Retrained one final model on the full training set using the best hyperparameters, then evaluated exactly once on the untouched 372-patient test set. Random Forest and SVM baselines were trained and evaluated on the identical train/test split, using raw concatenated features (no learned representation), to ensure a fair comparison.

Part 5: Results

Headline results (372 held-out test patients)

Model	Train accuracy	Test accuracy	Train–test gap	Test macro-F1	Misclassified
Supervised JISAE	100.0%	99.46%	0.54%	99.58%	2 / 372
Random Forest	100.0%	99.73%	0.27%	99.77%	1 / 372
SVM	99.4%	94.89%	4.51%	94.77%	19 / 372

Key findings

1. **Supervised JISAE clearly outperforms SVM and nearly matches Random Forest** — a substantial improvement over the original paper's unsupervised, post-hoc classification approach.
2. **The anticipated overfitting risk did not materialize.** Given the model's parameter count (~6.7 million) relative to the training set size (1,494 patients) and the relative ease of the classification task, overfitting was a reasonable concern going in. The train-test gap (0.54%) was smaller than SVM's (4.51%), suggesting the re-tuned hyperparameters provided sufficient implicit regularization.
3. **Per-class performance held up even for the smallest class** — melanoma (19 test samples) was classified with perfect precision and recall, ruling out the possibility that a good aggregate accuracy was masking poor performance on an underrepresented class.

Honest interpretation and limitation

Every method evaluated — including plain SVM — exceeded 94% accuracy, and Random Forest, using entirely raw, uncompressed features with no learned representation, was already near-ceiling at 99.73%. This indicates the six-class TCGA task offers limited discriminative difficulty: molecularly and histologically distinct cancer types are separable by nearly any reasonable classifier. The results therefore demonstrate that the architectural conversion is technically sound and genuinely competitive, but do not yet establish that JISAE's specific joint/individual embedding decomposition contributes value beyond what a simpler representation would achieve on this particular task. Distinguishing "the conversion works" from "this specific architecture matters" is the central open question motivating the next phase of the work.

Part 6: Current Status and Immediate Open Questions

- **Embedding-source ablation** (planned, code already supports it): testing classification performance using the joint embedding alone, the modality-specific embeddings alone, and all three combined, to directly test whether the joint/individual decomposition is functionally contributing to classification performance or whether the joint component alone suffices.
- **Joint-loss variant**: testing the MOSAE-style combined reconstruction-and-classification loss as a literature-backed alternative to the current phased approach, particularly relevant if a harder dataset reveals overfitting that the current approach does not exhibit on this easier task.
- **Harder dataset**: my advisor is sourcing a more challenging, less clearly separable classification task, since the current six-cancer-type problem has limited headroom to distinguish between methods.

Part 7: Future Direction — Interpretability

The next major phase of this project, discussed directly with my advisor, shifts focus from predictive performance toward **model interpretability**: developing methods to quantify variable importance within the supervised neural network, analogous to the variable importance measures naturally available from tree-based ensembles like Random Forest, but substantially harder to construct rigorously for deep architectures where predictive computation is distributed across many nonlinear layers rather than a sequence of interpretable splits.

This direction has a concrete motivating goal: producing a visualization or summary that can communicate, in an interpretable way, which genes or variables are driving the model's predictions at different stages of the network — giving an explanatory dimension to a model architecture that is otherwise a "black box." This is planned as the next literature review (specifically targeting variable-importance and explainability methods for classification neural networks), and there is a potential opportunity to develop this into a manuscript submission built around the visualization/interpretability angle specifically.

This future direction reflects what I now recognize as my core research interest: not simply building predictive models, but developing statistically principled methods for understanding *why* a model predicts what it does — particularly in domains where a prediction without an accompanying explanation has limited scientific or clinical value.

Part 8: Skills and Competencies Demonstrated

- **Deep learning implementation**: full PyTorch reimplementation of a published multi-branch autoencoder architecture, including custom loss functions and training loops
- **Literature synthesis**: structured review connecting a specific implementation question to established methodology across multiple published papers, correctly identifying the theoretical tradeoffs between competing approaches
- **Rigorous experimental design**: proper train/validation/test separation, stratified cross-validation, avoiding data leakage between hyperparameter selection and final evaluation
- **Hyperparameter optimization**: Bayesian search (Optuna) with appropriate objective specification and pruning
- **Critical self-evaluation**: identifying and clearly communicating the limitations of a strong result rather than overstating its significance
- **Reproducibility practices**: working from and extending a fully documented, version-controlled codebase; maintaining a clear audit trail from raw data through final results
- **Tooling and environment management**: resolving real-world engineering obstacles (Git LFS data retrieval, cross-platform GPU acceleration via Apple's MPS backend as a CUDA alternative, dependency management via conda environments)

Part 9: Related Work — Applied Machine Learning in Epidemiology

As a second, independent research experience, I contributed to a peer-reviewed study published in *Frontiers in Public Health* (Nahin & Hossen, 2026), examining socio-demographic and health-related determinants of pre-hypertension among Bangladeshi adults using nationally representative data from the 2022 Bangladesh Demographic and Health Survey (BDHS 2022, n = 10,981 after applying survey weights).

My role and contributions: software implementation, formal analysis, data curation, and visualization for the machine learning component of the study. Specifically, I implemented a Random Forest classification model to serve as a non-parametric robustness check against the study's primary survey-weighted logistic regression analysis, and to independently rank predictor variables by their contribution to classification performance via a variable importance analysis.

Key finding relevant to methodology: the Random Forest model's variable importance ranking (age, BMI, and sex as the top predictors) closely mirrored the independent findings from the logistic regression model, and its predictive performance (AUC = 0.619) was comparable to, rather than better than, the logistic regression model (AUC = 0.628). Rather than treating this as a disappointing result, we interpreted the convergence between two methodologically very different approaches as evidence strengthening confidence in the stability of the identified risk factors, and retained the more interpretable logistic regression model as the primary analytical method given the added complexity of the ensemble approach provided no compensating predictive benefit.

This experience — recognizing that a more flexible, higher-capacity model does not automatically outperform a simpler, more interpretable one, and treating that outcome as a meaningful methodological finding rather than a failure — directly shaped how I approached evaluating and reporting results in my thesis work, particularly the emphasis on stating clearly what a strong result does and does not prove.